

Higher order derivatives. Taylor expansion. Local extremum

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_i^2} = \partial_i^2 f \rightarrow \text{SECOND ORDER PARTIAL DERIVATIVE}$$

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial^2 f}{\partial x_i \partial x_j} = \partial_{i,j}^2 f \rightarrow \text{MIXED SECOND ORDER PARTIAL DERIVATIVE}$$

Ex: $f(x,y) = x^2y + (x+2y)^3$

$$\frac{\partial f}{\partial x} = 2xy + 3(x+2y)^2$$

$$\frac{\partial^2 f}{\partial x^2} = 2y + 6(x+2y)$$

$$\begin{aligned} &3(1+2y)^2 \\ &6(1+2y) \cdot (1+2y)' \\ &6 \cdot 2 \cdot (1+2y) \end{aligned}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (2xy + 3(x+2y)^2) = 2x + 12(x+2y)$$

$$\frac{\partial f}{\partial y} = x^2 + 6(x+2y)^2$$

$$\frac{\partial^2 f}{\partial y^2} = x^2 + 12 \cdot 2 \cdot (x+2y)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (x^2 + 6(x+2y)^2) = 2x + 6 \cdot 2 \cdot (x+2y)$$

Schwartz Theorem

If $f: A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ has **CONTINUOUS** second order partial derivatives $\xrightarrow{i \neq j}$

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Def: $f: A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$

gradient

$$H(x) = D^2 f(x) = D(\nabla f)(x) =$$

HESSIAN MATRIX

↓
SYMMETRIC if the second partial derivatives are CONT

$$\begin{bmatrix} \nabla \left(\frac{\partial f}{\partial x_1} \right) \\ \nabla \left(\frac{\partial f}{\partial x_2} \right) \\ \vdots \\ \nabla \left(\frac{\partial f}{\partial x_m} \right) \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_m \partial x_1} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_m \partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_m} & \dots & \dots & \frac{\partial^2 f}{\partial x_m \partial x_m} \end{bmatrix}$$

Taylor Theorem

$f: A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ of class C^2 $\leftarrow$ $\begin{matrix} \text{twice differentiable} \\ \text{continuous second order partial derivatives} \end{matrix}$
 $\downarrow$
 $x_0 \in A$.

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2} (x - x_0)^T \cdot H(x_0) (x - x_0) + \underbrace{R(x - x_0)}_{\text{remainder.}}$$

$$\frac{R(x - x_0)}{\|x - x_0\|^2} \rightarrow 0, \text{ as } x \rightarrow x_0$$

Taylor's expansion: $f(x_0 + h) = f(x_0) + \nabla f(x_0) \cdot h + \frac{1}{2} h^T \cdot H(x_0) h + R(h)$

$$\frac{R(h)}{\|h\|^2} \rightarrow 0$$

linear in h

quadratic in h

Prop: H an $m \times m$ matrix s.t. $x^T H x > 0$, $(\forall) x \in \mathbb{R}^m$?
 $\Rightarrow \exists H > 0$ s.t. $x^T H x \geq H \|x\|^2$, $(\forall) x \in \mathbb{R}^m$.

Corollary

$f: A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ of class C^2 $\leftarrow$ $\begin{matrix} \text{twice diff.} \\ \text{cont. second-order partial deriv.} \end{matrix}$

$$\nabla f(x_0) = 0$$

- If $h^T H(x_0) h > 0$, $(\forall) h \in \mathbb{R}^m \Rightarrow x_0$ local min.
- If $h^T H(x_0) h < 0$, $(\forall) h \in \mathbb{R}^m \Rightarrow x_0$ local max.

Def. An $m \times m$ matrix A is called:

- positive definite if $x^T A x > 0$, $(\forall) x \in \mathbb{R}^m$
- negative definite if $x^T A x < 0$, $(\forall) x \in \mathbb{R}^m$
- indefinite if $\exists x_1, x_2 \in \mathbb{R}^m$ s.t. $x_1^T A x_1 > 0 > x_2^T A x_2$

Prop. A symm. $m \times m$ matrix.

- A pos. definite $\Leftrightarrow$ positive eigenvalues
- A neg. definite $\Leftrightarrow$ negative eigenvalues
- A indef. $\Leftrightarrow$ both pos, neg. eigenvalues

Local extremum conditions

$f: A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ of class C^2

$x \in A$ critical point with $\nabla f(x) = 0$.

• If $H(x)$ pos. def. $\Rightarrow x$ local min.
(+ eigenvalues) 

• If $H(x)$ neg. def. $\Rightarrow x$ local max.
(- eigenvalues) 

• If $H(x)$ indef. $\Rightarrow x = \text{SADDLE POINT}$  (Pringle chip)

min. in some
directions

max. in other
directions

Ex: $f(x, y) = x^2 - y^2$

$$\nabla f(x, y) = (2x, -2y) = (0, 0) \Leftrightarrow (x, y) = (0, 0). \quad \text{SADDLE POINT}$$

$$H(0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

eigenvalues: 2, -2

